

WILLIAM ALEXAKIS

Athens, Greece

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EDUCATION

St. Catherine's British School

Athens, Greece

International Baccalaureate Diploma Programme

September 2026 — June 2028

Computer Science HL, Mathematics Analysis & Approaches HL, Physics HL

IGCSE Programme

September 2024 — June 2026

Computer Science, Mathematics Extended, Physics Triple Award

PROJECTS & RESEARCH

Greek vs English Degradation Under Consumer GGUF Quantization

May 2026 — August 2026

doi.org/10.5281/zenodo.22145680

- Wrote and published a solo-authored empirical NLP paper measuring how GGUF k-quantization degrades Greek more than English on Llama 3.2 3B and 1B, across 5 quantization levels, over a parallel FLORES+ corpus.
- Found that mean Greek-English KL divergence ratio never falls below 1.35 on either model; damage migrates from a tail of distorted tokens at high precision to typical tokens at low precision; and that Greek costs ~0.3 bits of effective precision compared to English.
- Established a practical usage floor of Q4_K_M for 3B and Q5_K_M for 1B for Greek-speaking users running Llama 3.2 on local inference, while Q2_K is unusable.

Phase Programming Language

August 2025 — December 2025

github.com/williamalexakis/phase

- Wrote a statically-typed programming language in C with zero dependencies in ~5,000 LOC, featuring a 25-opcode stack-based virtual machine, 5 primitive types, and tuple destructuring.
- Built a full compilation pipeline including a lexer, recursive-descent parser, static type checker, bytecode generator, and a stack-based virtual machine with scoped call frames.
- Implemented 23 error types with source-mapped diagnostics displaying exact file/line/column, visual markers, and diff-style fix suggestions.

ESSAYS

williamalexakis.github.io/essays/mathematics-of-scaled-dot-product-self-attention/

June 2026

williamalexakis.github.io/essays/chunk-based-file-reading-in-c/

June 2026

williamalexakis.github.io/essays/bytecode-interpretation-from-scratch-in-c/

December 2025

EXTRACURRICULAR ACTIVITIES

Open Source Contribution

June 2026 — Present

Contributor

- Implemented syntax error handler with source references for 6 error types in C++ for a programming language, reducing redundant code and improving error UX (github.com/omnimistic/algo-lang/pull/13).

StCatsHacks Line Robot Competition

December 2023 — December 2025

Team Leader & Lead Programmer

- Won 1st place for 3 consecutive years (2023 — 2025) leading a 3-person team.
- Designed a 3-sensor line-following algorithm to resolve simultaneous sensor activation on acute-angle turns, outpacing the standard PID approach used by competing teams on both speed and accuracy.

ADDITIONAL INFORMATION

Technical Skills

C, Python, Git

Spoken Languages

Fluent English, Native Greek